Equilibrium Transition Dynamics in Heterogeneous-Agent Economies

Abdoulaye Ndiaye using Claude Fable 5

July 20, 2026

Abstract

Quantitative work with heterogeneous-agent models routinely computes perfect-foresight transition paths: an interest-rate path clears the asset market at every date while the wealth distribution evolves from given initial conditions. We ask when such equilibrium paths exist. In the continuous-time Huggett economy with CRRA utility we show that market clearing stated in *stocks* is a degenerate restriction on the rate path: today's interest rate has no contemporaneous effect on today's aggregate wealth. The operative form of clearing is a *flow* condition—aggregate consumption equals aggregate income plus interest on the bond supply—in which the rate's only instantaneous leverage comes from the bond stock and from hand-to-mouth households, whose interest burden moves one-for-one with the rate. We prove that equilibrium transitions exist and are unique over short horizons whenever the debt of hand-to-mouth households is smaller than the bond supply, and that whether households remain at the borrowing constraint near the end of a transition is governed by terminal conditions rather than by discounting. With risk aversion above one, monotonicity of market clearing in the rate path fails on known economies, and we argue that multiple self-fulfilling transition paths connecting the same endpoints become a live possibility.

JEL codes: C62, D52, E21, D31. *Keywords:* incomplete markets, borrowing constraints, transition dynamics, existence of equilibrium, mean field games.

1 Introduction

Modern quantitative macroeconomics runs on transition paths. To evaluate a credit crunch, a fiscal reform, or a monetary shock in a Bewley–Huggett–Aiyagari economy, one computes the perfect-foresight response: starting from an initial distribution of wealth, households optimize against a conjectured path of prices, the wealth distribution evolves under their decisions, and the price path is adjusted until markets clear at every date (Kaplan, Moll, and Violante, 2018; Boppart, Krusell, and Mitman, 2018; Auclert et al., 2021; Achdou et al., 2022). The algorithms that do this are by now standard. The theory behind them is not: while existence and (under conditions) uniqueness of *stationary* equilibria are well understood (Aiyagari, 1994; Açıkgöz, 2018; Achdou et al., 2022; Höfer, 2025), there is, to our knowledge, no theorem guaranteeing that the equilibrium transition path being computed exists, for any horizon, any preference parameters, or any level of asset supply. This paper studies that question in the canonical laboratory: the continuous-time Huggett economy with CRRA utility, two income states, a borrowing constraint, and a bond in fixed supply $B \geq 0$, over a horizon $[0, T]$.

Why is this not a routine extension of the steady-state theory? In a stationary equilibrium the entire influence of the wealth distribution on any single household is summarized by one number, the interest rate r. Fixing r decouples the household problem from the distribution, and existence reduces to finding a root of a scalar excess-supply function—Aiyagari's celebrated diagram, formalized by an intermediate-value argument. Along a transition, the unknown is a *path* $r(\cdot)$, pinned down by a clearing condition at a continuum of dates. Three economic facts, which we formalize, make this fixed-point problem behave very differently from its scalar ancestor.

First, almost nobody responds to today's interest rate today. A household's consumption at date t is determined by its wealth and by the price path it expects *from t onward*; changing r at the single instant t leaves its date-t behavior unchanged. The wealth distribution at t, in turn, depends on prices *before* t. Consequently today's rate has no contemporaneous effect on today's aggregate wealth: market clearing stated in stocks ("aggregate wealth equals bond supply at every date") is a degenerate restriction on the rate path—in technical language, a first-kind integral equation, which admits no stable inversion (Theorem 4.3). The exception, and it is the crux of the paper, is a group whose behavior *does* depend on today's rate: hand-to-mouth households at the borrowing constraint, who consume their income flow $y_1 + r(t)\underline{a}$ and whose interest burden therefore moves one-for-one with $r(t)$; and bondholders in the aggregate, whose interest income moves with $r(t)B$.

Second, the operative form of market clearing is a flow condition. Differentiating aggregate wealth along the distributional dynamics shows that clearing at every date is equivalent to clearing at the initial date—a restriction on the *data*, not on prices—plus the aggregate budget identity

$$C(t) \;=\; Y(t) + r(t)B \qquad \text{at every } t,$$

aggregate consumption equals aggregate income plus interest on the bond supply (Lemma 4.2). In this form the rate has exactly the two contemporaneous levers just described: the sensitivity of excess demand to $r(t)$ is $B + |\underline{a}|\, m_1(t)$, where $m_1(t)$ is the mass of constrained households. All of our existence results are built on this diagonal. It is telling that numerical practice discovered the same structure by trial: the transition algorithm of Achdou et al. (2022, §5.5) updates the rate path using the *time derivative* of excess supply—the flow—rather than its level.

Third, the borrowing constraint itself becomes a moving part. In the stationary economy, impatience ($r < \rho$) guarantees that the constraint binds and that a point mass of hand-to-mouth households forms at the borrowing limit. Along a transition this is no longer governed by discounting. We show that whether households stay at the constraint near the end of the horizon is controlled by the *terminal condition*: with the standard choice—continuation values from a stationary equilibrium at rate r^*—the constrained group stays put at the end of the horizon if and only if $r(T) \geq r^*$, and disperses whenever the path ends below r^*, even if $r(t) < \rho$ throughout (Corollary 3.3). The economics is elementary once seen: constrained households are debtors, so a rate *below* the one embedded in their continuation values lowers their debt service, leaves them income to spare, and they save their way off the constraint. Since the constrained mass $m_1(t)$ is also the rate's main lever on market clearing, the model's two degeneracies—at the borrowing limit and in the price—are coupled: episodes in which the constrained group disperses are episodes in which the market loses its contemporaneous handle on the interest rate.

Results. With this structure in place, our main results are as follows.

(1) *Existence and uniqueness for short transitions with positive bond supply* (Theorem 5.1). If the initial distribution clears the market, if the data imply an interior initial rate (an "anchoring" condition), and if

$$\theta \;:=\; \frac{|\underline{a}| \cdot \sup_t m_1(t)}{B} \;<\; 1,$$

then an equilibrium transition exists and is unique on $[0, T]$ for T below an explicit threshold. The condition $\theta < 1$ has a transparent reading: a rise dr in the rate mechanically raises the consumption the economy must absorb by $B\,dr$ (interest paid on the bond supply) and lowers hand-to-mouth consumption by $|\underline{a}|m_1\,dr$ (interest charged on constrained debt); the equilibrium feedback loop is stable when the second effect is smaller than the first—equivalently, when the debt of hand-to-mouth households is smaller than the bond supply. That is the stability condition of the transition.

(2) *Zero net supply lives or dies by the constrained mass* (Theorem 5.2, Proposition 5.3). With $B = 0$, the flow condition pins aggregate consumption to aggregate income, and the *only* contemporaneous lever of the rate is the constrained group. If the initial distribution carries a point mass at the constraint and the constraint remains binding, that mass decays no faster than the rate at which its members draw the high income state, and a short-horizon equilibrium exists. If instead $B = 0$ and no one is constrained on some time interval, market clearing there contains today's rate *nowhere*: the constraint becomes a restriction linking only past and future, no local solution theory applies, and—consistent with the nonexistence results of Ambrose (2021) for a frictionless relative of this model—exact clearing is generically the wrong demand. Relaxations (a slightly elastic bond supply, or a price rule responding to excess demand) restore a well-posed problem; we make precise the sense in which they are a necessity rather than a computational convenience.

(3) *The household problem along a price path is harder than its stationary counterpart, in identifiable places* (Section 3, Appendices A and B). We provide a complete well-posedness package: the household's value function is the unique constrained viscosity solution of its Bellman equation along any admissible rate path; marginal values admit two-sided bounds, constructed by explicit barrier arguments because the stationary shortcut—the binding constraint pinning the boundary gradient—is unavailable; consumption near the constraint follows a square-root law whose speed coefficient $\nu_1(t)$ acquires a new term $\dot{r}(t)\underline{a}$ (a *rising* rate slows the approach to the constraint through the debtor's income channel); and the wealth distribution evolves as a measure with a point mass at the constraint that grows while the constraint binds and detaches *discontinuously* when it stops binding. Two features of the finite horizon are genuinely simpler and worth recording: neither $\rho > 0$ nor $r < \rho$ plays any role in household well-posedness.

(4) *Long transitions bifurcate on risk aversion* (Section 6). For $\gamma \leq 1$ (intertemporal elasticity of substitution at least one—the range in which the stationary equilibrium is known to be unique), a natural monotonicity property of aggregate consumption in the rate path, if it holds, localizes all equilibria between two computable paths and yields uniqueness under a verifiable collapse condition (Theorem 6.2); we explain exactly why this monotonicity is a hypothesis rather than a theorem in finite horizon. For $\gamma > 1$ it is not available even as a hypothesis: Höfer (2025) constructs economies whose stationary aggregate saving *decreases* in the interest rate over a range, and on long horizons such economies produce singular linearizations of the market-clearing map and, plausibly, multiple self-fulfilling transition paths connecting the same endpoints—anticipation-driven loops in which a believed future path of rates alters consumption today, reshapes the wealth distribution, and validates the belief (Proposition 6.3). For risk aversion above one—the empirically common calibration—uniqueness of transition dynamics should be regarded as an assumption of computational practice, not a theorem.

1.1 Related literature

The stationary theory is by now well developed. Achdou et al. (2022) formulate the continuous-time system, prove existence of stationary equilibria by the Aiyagari intermediate-value argument, and prove uniqueness when the intertemporal elasticity of substitution is at least one and borrowing is prohibited; Açıkgöz (2018) rigorizes the discrete-time counterpart and gives a calibrated multiplicity example. Mathematically complete stationary treatments are given by Shigeta (2023) (bounded utility) and Höfer (2025) (CRRA utility under a no-borrowing limit; ergodicity of the wealth process, continuity of aggregate assets in r, and the non-monotonicity construction for $\gamma > 1$ used in Section 6); Bayer, Rendall, and Wälde (2019) analyze the invariant wealth distribution at fixed prices. All of these concern steady states.

For transitions, the closest prior work is Ambrose (2021), who studies the household-wealth model of Achdou et al. (2014)—a variant with diffusive income, *no* borrowing constraint (compactly supported wealth densities), and zero net supply. He proves small-time existence and uniqueness for a *relaxed* problem in which exact clearing is replaced by constancy of the flow imbalance, and gives a nonexistence result (his §8) for the exact problem: with generic data, no exactly-clearing solution exists over short horizons. In our language, his model has $B = 0$ and, by construction, no mass at a borrowing constraint, so the rate's contemporaneous lever on clearing is identically zero; his Remark 1 notes explicitly that mass at a wealth boundary would contribute "another term proportional to r" through which clearing could be enforced. The present paper can be read as the constrained-economy counterpart: the point mass of hand-to-mouth households (and positive bond supply) restores exactly that missing term, and with it an existence theory for exact clearing. Chen et al. (2025) obtain a time-dependent market-clearing rate path in an overlapping-generations economy, under a never-binding natural borrowing limit and a short-lifespan restriction that plays the role of our short horizon.

Methodologically the paper sits in the mean field games (MFG) literature. Finite-horizon well-posedness results exist for price-mediated MFG in which the price is a multiplier on a constraint involving agents' *controls* (Gomes and Saúde, 2021; Fujii and Takahashi, 2022), for oligopoly models with prices that are explicit functionals of the distribution (Graber and Bensoussan, 2018), for MFG of controls (Cardaliaguet and Lehalle, 2018; Kobeissi, 2022), including a state-constrained case under a smallness condition on the dependence of preferences on the joint law of states and controls (Graber and Mayorga, 2021)—a condition that, in our multiplier formulation, scales like $1/B$ and fails precisely in the economically central small-B regime— and for general non-separable forward–backward parabolic systems over short horizons (Cirant, Gianni, and Mannucci, 2020). In all of these, either the clearing constraint responds nondegenerately to the contemporaneous price through the controls, or there is no exact clearing constraint at all; none features the combination present here—a hard borrowing constraint with a boundary point mass, switching income risk, a Hamiltonian singular at zero marginal value, and a price path pinned by instant-by-instant clearing in stocks. The deterministic theory of MFG with state constraints (Cannarsa and Capuani, 2018; Cannarsa, Capuani, and Cardaliaguet, 2021) supplies the right language for constrained trajectories; the planning problem (Porretta, 2014) treats a single terminal constraint enforced by a multiplier, where our clearing imposes one scalar constraint per instant.

Finally, our degeneracy theorem speaks to computational practice. In discrete time, the sequence-space Jacobian of Auclert et al. (2021) is precisely a discretization of the linearized market-clearing map that our Theorem 4.3 analyzes; the theorem implies that in the continuous-time limit the Jacobian of *stock*-level clearing has vanishing diagonal and is compact—ill-conditioned at fine time discretizations—while the flow form retains an invertible diagonal $B + |\underline{a}|m_1(t)$. This is consistent with, and explains, the practice of updating rate paths via flow imbalances (Achdou et al., 2022, §5.5).

The paper proceeds as follows. Section 2 lays out the economy. Section 3 analyzes the household problem along a rate path, with the binding/dispersal results. Section 4 develops the structure of market clearing. Section 5 contains the short-horizon existence and uniqueness theory, Section 6 the long-horizon analysis. Section 7 concludes. Appendices A –C contain the technical package and proof architectures.

2 The economy

2.1 Households

Time is continuous over a finite horizon $[0, T]$. There is a continuum of households with CRRA preferences

$$\mathbb{E}_0 \int_0^\infty e^{-\rho t}\, u(c_t)\, dt, \qquad u(c) = \frac{c^{1-\gamma}}{1-\gamma},\ \gamma > 0 \quad (u = \log \text{ for } \gamma = 1), \qquad \rho \geq 0.$$

A household's income y_t takes two values $0 < y_1 < y_2$ ("unemployed"/"employed"), switching with Poisson intensities $\lambda_1, \lambda_2 > 0$. Wealth is held in a riskless bond and evolves as

$$\dot{a}_t = y_t + r(t)\, a_t - c_t, \qquad a_t \;\geq\; \underline{a}, \qquad -\infty < \underline{a} < 0,$$

where $r(t)$ is the market interest rate, taken as given by the household, and $\underline{a}$ is an exogenous borrowing limit. The horizon is closed by a terminal valuation $v_{T,j}(a)$ of end-of-horizon wealth; the canonical choice, following Achdou et al. (2022), is the value function $v_\infty(\cdot\,; r^*)$ of a stationary equilibrium with rate r^*, so that the finite-horizon problem is a truncation of a transition converging to a steady state.

The household's value functions $v_j(a, t)$, $j = 1, 2$, solve the backward Bellman (HJB) system

$$\rho v_j = \partial_t v_j + \max_{c \geq 0} \Big\{ u(c) + \partial_a v_j \left(y_j + r(t)a - c \right) \Big\} + \lambda_j \left(v_{-j} - v_j \right) \qquad \text{on } (\underline{a}, \infty) \times (0, T), \quad (1)$$

with terminal condition $v_j(\cdot,T) = v_{T,j}$ and the *state-constraint* boundary condition

$$\partial_a v_j(\underline{a},t) \;\geq\; u'\big(y_j + r(t)\underline{a}\big), \qquad j = 1,2, \tag{2}$$

which says that the marginal value of wealth at the borrowing limit is at least the marginal utility of the consumption level $\underline{c}_j(t) := y_j + r(t)\underline{a}$ that just keeps the household at the limit; equivalently, that chosen saving at the limit is nonnegative, so the constraint is never violated. Optimal consumption satisfies $c_j = (u')^{-1}(\partial_a v_j)$ in the interior and $c_j(\underline{a},t) = \min\{(u')^{-1}(\partial_a v_j(\underline{a}^+,t)),\, \underline{c}_j(t)\}$ at the limit; the saving policy is $s_j(a,t) = y_j + r(t)a - c_j(a,t)$.

The cross-sectional distribution of (income, wealth) is described by measures $G_j(\cdot,t)$ on $[\underline{a},\infty)$ of total mass one, evolving under the households' saving policies and the income switching by the Kolmogorov forward equation, understood in a weak (measure) sense that accommodates a point mass $m_1(t)$ of households *at* the borrowing limit (Definition B.1 in Appendix B); $G_j(\cdot,0) = G_{0,j}$ is given. This system—equations (12)–(17) of Achdou et al. (2022), together with their market-clearing condition (4)—is the transition-dynamics version of the continuous-time Huggett model (Huggett, 1993).

2.2 Equilibrium

Definition 2.1. Given data (G_0, v_T) and bond supply $B \in [0,\infty)$, an *equilibrium transition* on $[0,T]$ is a continuous interest-rate path $r(\cdot)$, household value functions solving (1)–(2) with terminal data v_T (in the constrained viscosity sense, Definition A.2), and distributions solving the forward equation from G_0 under the induced policies, such that the bond market clears at every date:

$$S(t) \;:=\; \sum_{j=1,2} \int_{[\underline{a},\infty)} a\, dG_j(a,t) \;=\; B \qquad \text{for all } t \in [0,T]. \tag{3}$$

Throughout, candidate rate paths lie in a band $\mathcal{K} = \{r \in C([0,T]) : r_{\min} \leq r \leq r_{\max}\}$ with

$$r_{\max} < \frac{y_1}{|\underline{a}|}, \tag{4}$$

which guarantees $\underline{c}_1(t) = y_1 + r(t)\underline{a} \geq \underline{c} > 0$: even the low-income household at the borrowing limit can service its debt out of income with something left to consume. Condition (4) is the transition version of the requirement that the borrowing limit be tighter than the natural limit; it is what makes the constraint economically meaningful (households at the limit are hand-to-mouth, not defaulting) and, mathematically, it is the inward-pointing condition on which the entire boundary analysis rests. The full list of maintained assumptions on the data (G_0, v_T) is collected in Appendix A (Assumptions 1–4, covering primitives, the rate band, and the data); the terminal valuation is concave, increasing, with marginal values bounded above and decaying no faster than CRRA marginal utility of a linearly growing consumption level—properties satisfied by the canonical choice $v_\infty(\cdot\,; r^*)$.

2.3 Why transitions are not steady states strung together

It is worth previewing, in words, where the difficulty lies, because it is *not* where the steady-state theory suggests.

In a stationary equilibrium the logic is: fix r, solve the household problem, compute the implied stationary wealth stock $S_{\text{stat}}(r)$, and find r with $S_{\text{stat}}(r) = B$. The map $r \mapsto S_{\text{stat}}(r)$ is a well-behaved scalar function, and existence follows from its boundary behavior (Aiyagari, 1994; Achdou et al., 2022). One might hope to treat a transition as a family of such problems indexed by t. This fails, for an economic reason: $S(t)$ is a *stock*, accumulated from past saving decisions, each of which was made looking at *future* rates. The rate at date t influences wealth at date t not at all; it influences wealth *after* t through household budgets, and consumption *before* t through anticipation. So the natural analogue of Aiyagari's diagram—move $r(t)$ until $S(t)$ hits B—moves nothing. Section 4 makes this precise and locates the two places where the contemporaneous rate does bite: interest on the bond supply, and the budgets of constrained households. Everything constructive in this paper is built on those two levers; and the second lever is itself an equilibrium object, switched on and off by the binding analysis of the next section.

3 Household behavior along an interest-rate path

This section takes the rate path $r \in \mathcal{K}$ as given and establishes what we need about individual behavior: that the household problem is well posed at all (Theorem 3.1), and—the economically substantive part—when households at the borrowing limit stay there (Section 3.1). Readers willing to take well-posedness on faith can skim to Section 3.1; the technical package, including all regularity and stability estimates used later, is developed in Appendix A.

Theorem 3.1 (the household problem is well posed)**.** *Under the maintained assumptions (Appendix A), for every $r \in \mathcal{K}$ the household's value function is the unique constrained viscosity solution of (1)–(2) with terminal data v_T, in the class of continuous functions concave and increasing in wealth with linear growth. It is continuously differentiable in wealth on $(\underline{a}, \infty)$, with a one-sided derivative at $\underline{a}$; consumption is continuous and nondecreasing in wealth; marginal values admit two-sided bounds, uniform over $\mathcal{K}$,*

$$0 < \kappa\, e^{-CT}(1 + a - \underline{a})^{-\gamma} \;\le\; \partial_a v_j(a,t) \;\le\; \max\{\bar{p}_T,\, u'(\underline{c})\}\, e^{(r_{\max}-\rho)^+ T},$$

where $\bar{p}_T$ bounds the terminal marginal value; equivalently, consumption is bounded below by a positive constant and grows at most linearly in wealth. The value function and the policies *depend continuously—and the value Lipschitz-continuously—on the rate path, with only the* future *of the path entering (Lemma A.8).*

Three comments on the economics of the proof (details in Appendix A). First, the bounds on marginal value are where finite horizon differs from the steady state. Stationarily, the binding constraint supplies the boundary marginal value *exactly*—$\partial_a v_1(\underline{a}) = u'(y_1 + r\underline{a})$ whenever $r < \rho$—and concavity does the rest. Along a transition, binding is not guaranteed (next subsection), and the bounds must be built by comparison with explicit sub- and super-solutions: the upper bound propagates backward the larger of the terminal marginal value and the boundary obstacle $u'(\underline{c})$; the lower bound is a CRRA-shaped barrier expressing that consumption cannot outgrow wealth linearly. The cap $\bar{p}_T$ on terminal marginal values is a genuine assumption on the terminal data, not a free good. Second, the bounds keep marginal values in a compact positive range, which is what tames the CRRA Hamiltonian $H(p) = \max_c\{u(c) - pc\}$, whose slope $H'(p) = -p^{-1/\gamma}$ blows up as $p \downarrow 0$ (marginal values near zero correspond to unboundedly rich households). Third—an honest simplification—neither $\rho > 0$ nor $r < \rho$ appears anywhere: on a finite horizon the terminal valuation replaces transversality, and impatience plays no coercive role. The conditions involving ρ reappear only through the terminal data, which is exactly where the next subsection locates them.

3.1 When does the borrowing constraint bind?

Say the constraint *binds* at date t if households at the limit stay there: $s_1(\underline{a}, t) = 0$, equivalently $\partial_a v_1(\underline{a}^+, t) = u'(\underline{c}_1(t))$. When it binds, the point mass $m_1(t)$ of hand-to-mouth households persists and grows; when it fails ($s_1(\underline{a}, t) > 0$), the constrained group saves its way off the limit and the mass disperses. In the stationary economy binding is automatic under impatience, $r < \rho$. Along a transition the governing comparison is different:

Lemma 3.2 (binding near the end of the horizon)**.** *(i) If the terminal marginal value at the limit exceeds the marginal utility of hand-to-mouth consumption at the terminal rate—$\partial_a v_{T,1}(\underline{a}) > u'(y_1 + r(T)\underline{a})$—then the constraint does* not *bind on an interval $(T - \delta, T)$, regardless of how $r(t)$ compares to ρ. (ii) If the reverse strict inequality holds, the constraint binds near T.*

Corollary 3.3 (the knife edge at the canonical terminal data)**.** *Let $v_T = v_\infty(\cdot\,; r^*)$, the stationary value at rate r^*, so that $\partial_a v_{T,1}(\underline{a}) = u'(y_1 + r^*\underline{a})$ by the stationary binding equality. Then the boundary saving rate at the end of the horizon is*

$$s_1(\underline{a}, T^-) \;=\; \big[(r(T) - r^*)\,\underline{a}\big]^+,$$

and since $\underline{a} < 0$: the constraint binds at T^- if and only if $r(T) \geq r^*$. *Whenever the path ends below the rate embedded in the continuation values, the hand-to-mouth group disperses near the* *end of the transition—even if $r(t) < \rho$ throughout.*

The economics bears repeating because it inverts a steady-state intuition. Households at the borrowing limit are *debtors*: their consumption is income net of debt service, $y_1 + r(t)\underline{a}$, which *falls* as the rate rises. Their continuation plan, embedded in v_T, was drawn at rate r^*. If the transition delivers a terminal rate below r^*, debt service is lighter than the continuation plan assumed; the constrained household finds itself with slack, saves, and detaches from the limit. Impatience (ρ) never enters: over a short remaining horizon, behavior is governed by the terminal valuation, not by discounting. A closed-form illustration with no income risk, in which binding fails on an entire terminal interval despite $r < \rho$, is given in Example A.7 (Appendix A); the same example exhibits the turnpike property—binding is restored at dates far from T, where the stationary logic reasserts itself.

Near the constraint, consumption follows a square-root law, as in the stationary analysis of Achdou et al. (2022), but with a time-dependent speed:

Lemma 3.4 (square-root law along a transition)**.** *Let $r \in C^1$ and suppose the constraint binds on $[t_0, t_1] \subset (0, T)$. Define*

$$\nu_1(t) \;:=\; \frac{(\rho + \lambda_1 - r(t))\, u'(\underline{c}_1(t)) - \lambda_1\, \partial_a v_2(\underline{a}, t)}{-u''(\underline{c}_1(t))} \;+\; \dot r(t)\, \underline{a}.$$

If $\nu_1 \geq \underline{\nu} > 0$ on $[t_0, t_1]$, then locally uniformly there

$$c_1(a,t) = \underline{c}_1(t) + \sqrt{2\nu_1(t)}\,\sqrt{a - \underline{a}} + O(a - \underline{a}), \qquad s_1(a,t) \sim -\sqrt{2\nu_1(t)}\,\sqrt{a - \underline{a}}.$$

Relative to the stationary coefficient, time dependence adds the term $\dot r(t)\,\underline{a}$: since $\underline{a} < 0$, a *rising* rate path lowers ν_1—anticipated increases in debt service slow the approach to the constraint—and $\nu_1(t) \downarrow 0$ is precisely the incipient dispersal regime of Corollary 3.3. The formal expansion and its rigorous version are discussed in Appendix A.

Finally, the constrained mass itself. While the constraint binds, $m_1(t)$ grows by the inflow of low-income households hitting the limit and shrinks only as its members draw the high income state (at rate λ_1). When binding fails, the entire mass detaches *at once* and moves into the interior as a traveling atom: $m_1(t)$ drops to zero discontinuously (Proposition B.4, Appendix B). This bookkeeping matters beyond descriptive interest: $m_1(t)$ is exactly the market's contemporaneous lever on the interest rate, so the binding analysis of this subsection switches the tractability of the equilibrium problem on and off.

4 Market clearing along a transition

We now turn to equilibrium. Throughout, $S(t)$ denotes aggregate wealth (3), $C(t) = \sum_j \int c_j \, dG_{j,t}$ aggregate consumption, and $Y(t) = \sum_j \int y_j \, dG_{j,t}$ aggregate income.

4.1 Compatibility and the flow form of clearing

Proposition 4.1 (the data must clear the market at date zero)**.** *If an equilibrium transition exists, then* $S(0) = \sum_j \int a\, dG_{0,j} = B$*. No choice of prices can repair incompatible initial data: clearing at* $t = 0$ *involves the initial distribution alone.*

This is obvious once stated, but it is a substantive discipline on policy experiments: an "MIT shock" that revalues wealth or bonds at $t = 0$ must do so consistently with (3), or no equilibrium path exists at all.

Lemma 4.2 (clearing in stocks equals clearing at date zero plus clearing in flows)**.** *Along any solution of the distributional dynamics with uniformly bounded first moments,* $t \mapsto S(t)$ *is Lipschitz and*

$$\dot{S}(t) \;=\; \sum_j \int s_j \, dG_{j,t} \;=\; Y(t) + r(t)S(t) - C(t),$$

the boundary contributing nothing (households at the limit have zero saving while the constraint binds; dispersing mass is counted in the interior). Consequently, for $r \in \mathcal{K}$*:*

$$S(t) = B \quad \forall t \in [0,T] \qquad \Longleftrightarrow \qquad S(0) = B \quad \text{and} \quad C(t) = Y(t) + r(t)B \quad \forall t \in [0,T].$$

The flow form is the economy's aggregate budget identity: consumption absorbs income plus the interest paid on the bond supply. For $B = 0$ it says aggregate consumption is pinned to aggregate income at every date—an exogenous restriction in which, as we will see, the interest rate barely appears.

4.2 Stock-level clearing is degenerate

Theorem 4.3 (no contemporaneous leverage in stocks)**.** *Fix data and a path* $r \in \mathcal{K}$*, and consider the equilibrium map* $r \mapsto S(\cdot\,; r)$*.*

(i) Pinning. $S(0; r)$ *is independent of* r*: near* $t = 0$ *the stock condition restricts only the data.*

(ii) Vanishing diagonal. *For path perturbations* δr *supported on a window* $(t_0 - h, t_0 + h)$*,*

$$|S(t_0; r + \delta r) - S(t_0; r)| \;\leq\; C\, h\, (1 + T)\, \|\delta r\|_\infty$$

(under the bounded-density regime (D1); with an additional factor $1 + \log_+\big(1/(h\|\delta r\|_\infty)\big)$ *in the realistic boundary-layer regime (D1*$'$*)—either way the bound vanishes with* h*): the rate at date* t_0 *has no contemporaneous effect on wealth at date* t_0*.*

(iii) Compactness. *The linearized map* $DS(\cdot\,; r)$ *is a compact integral operator on* $C([0,T])$*; the equation* $S(\cdot\,; r) = B$ *is a first-kind equation, admitting no bounded local inverse and*

no Picard or Newton theory, and a stock-level tâtonnement $r \mapsto r + \kappa(S - B)$ cannot be closed: unlike the stationary problem, sign information on excess supply at the edges of the rate band is unavailable pointwise in t, because $S(t;\cdot)$ is a functional of the entire path.

Any existence argument must therefore run through the flow form of Lemma 4.2.

The economic content of (ii) is the observation of the introduction: consumption at t is forward-looking, wealth at t is backward-looking, and the instant t itself has measure zero in both. The exception is hidden in the constant C: it is built from the *integrated* response of consumption and savings to the rate window, which includes the mechanical response of constrained households. That response reappears, no longer averaged away, in the flow form: differentiating the flow-clearing residual with respect to the contemporaneous rate,

$$\frac{\partial}{\partial r(t)}\Big[C(t) - Y(t) - r(t)B\Big] \;=\; m_1(t)\,\underline{a} \;-\; B \;=\; -\big(B + |\underline{a}|\,m_1(t)\big), \tag{5}$$

because the only households whose date-t consumption depends on the date-t rate are those at the borrowing limit, consuming $y_1 + r(t)\underline{a}$. The flow constraint has a uniformly invertible diagonal if and only if $B + |\underline{a}|m_1(t)$ is bounded away from zero—automatically for $B > 0$; through the constrained mass, hence through the binding analysis of Section 3.1, when $B = 0$.

Theorem 4.3 also rationalizes numerical practice. The transition algorithm of Achdou et al. (2022, §5.5) updates $r^{\ell+1}(t) = r^{\ell}(t) - \xi(t)\,\frac{d}{dt}S^{\ell}(t)$: the update signal is the *flow* imbalance, the object with a nondegenerate diagonal, not the excess stock. Likewise, in discrete-time sequence-space methods (Auclert et al., 2021), our result implies that the Jacobian of stock-level clearing becomes ill-conditioned as the period shrinks, while flow-level clearing does not.

5 Short transitions: existence and uniqueness

We now solve the equilibrium problem over short horizons, first with positive bond supply, then at zero net supply. Throughout: $S(0) = B$ (Proposition 4.1), the maintained assumptions, and the household package of Section 3. Two density regimes for the wealth distribution near the constraint are distinguished: (D1), densities bounded near $\underline{a}$, uniformly in t and over the band $\mathcal{K}$; and (D1′), the realistic regime in which the density blows up like $(a - \underline{a})^{-1/2}$ with a constant uniform in t and over $\mathcal{K}$—the profile generated by the square-root consumption law.

5.1 Positive bond supply: the multiplier map and the hand-to-mouth threshold

For $B > 0$, Lemma 4.2 converts equilibrium into a fixed point over rate paths: $r = \Psi(r)$ with

$$\Psi(r)(t) \;:=\; \frac{C(t;r) - Y(t;r)}{B},$$

the rate that makes bond interest absorb the excess of consumption over income implied by the path r. The map Ψ is compact (its range is uniformly bounded and equicontinuous—there is no identity part, unlike a stock-level tâtonnement), and its Lipschitz constant decomposes along economic channels: for paths r, r' with $\eta = \|r - r'\|_\infty$,

$$\|\Psi(r) - \Psi(r')\|_\infty \le \underbrace{\theta\,\eta}_{\substack{\text{contemporaneous:}\\ \text{hand-to-mouth budgets}}} + \underbrace{C_1 T\,\eta}_{\substack{\text{anticipation:}\\ \text{policies respond to future rates}}} + \underbrace{C_2 T\,\eta\left(1 + \log_+(1/\eta)\right)}_{\substack{\text{history:}\\ \text{distribution shaped by past rates}}}, \qquad \theta := \frac{|\underline{a}|\ \text{su}}{\cdots}$$

with the logarithm absent under (D1). The anticipation and history channels are weak over short horizons—each carries a factor T—but the contemporaneous channel is not: it is the hand-to-mouth budget response, present at any horizon, and it is governed by θ, the ratio of constrained households' debt exposure to the bond supply. One further data condition is needed to keep the fixed point inside the rate band: as $T \to 0$ the map Ψ converges to an affine map of the instantaneous rate (the affine term again being the hand-to-mouth budget), whose fixed point r^{**} solves a scalar equation in the data; we require r^{**} to be interior to the band (condition (ANCH), Appendix C).

Theorem 5.1 (existence and uniqueness for short transitions, $B > 0$)**.** *Let $B > 0$, $S(0) = B$, (ANCH), and*

$$\theta \;\le\; \bar\theta \;<\; 1 \qquad \text{(hand-to-mouth debt exposure below bond supply).}$$

Then there is $T^* > 0$, depending on the data, $\bar\theta$, and the anchoring margin, such that for $T \le T^*$:

(i) an equilibrium transition exists;

(ii) under (D1), the map Ψ is a contraction with modulus $\bar\theta + CT < 1$: the equilibrium is unique in the band, and the natural iteration—guess a rate path, solve households backward, aggregate forward, update the rate from the flow-clearing condition—converges geometrically from any starting path;

(iii) under (D1′), any two equilibria are within $\exp\big(-(1 - \bar\theta - (C_1 + C_2)T)/(C_2 T)\big)$ of each other: uniqueness holds up to an ambiguity exponentially small in $1/T$.

The stability condition $\theta < 1$ deserves emphasis; it is, to our knowledge, new. Consider a contemporaneous increase dr in the rate. Flow clearing requires the economy to absorb $B\,dr$ of additional interest income as consumption. But the same increase *taxes* hand-to-mouth debtors, whose consumption falls mechanically by $|\underline{a}|\,m_1\,dr$. If the second effect exceeds the first, the market-clearing response to excess consumption is to *raise* required consumption further: the multiplier loop is unstable, and the fixed-point argument fails. The condition is economically interpretable and, in principle, measurable: the debt of constrained households relative to the outstanding stock of liquid assets. Economies with large hand-to-mouth populations, deep borrowing, and thin asset markets are the ones in which transition equilibria are fragile.

A natural question is whether the threshold is an artifact of the fixed-point scheme. It is not, in the following sense: one can solve the flow-clearing condition *exactly* for $r(t)$ (the constrained group's budget is known in closed form), eliminating the contemporaneous channel altogether; but the resulting map depends on the constrained mass $m_1(t;r)$ *separated* from the rest of the distribution, and the mass alone is only $\frac{1}{2}$-Hölder stable in the rate path under (D1′)—mass exchanged between the atom and the adjacent boundary layer moves the aggregate only at second order, but moves the atom at first order. Under (D1) the elimination works and $\theta < 1$ can be dispensed with; under the realistic (D1′) the threshold is load-bearing (Proposition C.1, Appendix C).

5.2 Zero net supply: the constrained mass is the market

At $B = 0$ the flow condition reads $C(t) = Y(t)$: aggregate consumption is pinned to aggregate income, and by (5) the *only* contemporaneous lever of the rate on clearing is the hand-to-mouth group, with weight $|\underline{a}|\, m_1(t)$.

Theorem 5.2 ($B = 0$ with a constrained population)**.** *Let $B = 0$, $S(0) = 0$, $m_1(0) > 0$, and assume—as an explicit hypothesis—that the constraint binds along the band throughout $[0, T]$. (Necessary for this is the terminal-data compatibility $\partial_a v_{T,1}(\underline{a}) \le u'(y_1 + r_{\min}\underline{a})$, which for canonical terminal data reads $r_{\min} \ge r^*$ by Corollary 3.3; but terminal compatibility controls binding only near T, and interior dispersal episodes along dipping-and-rebounding paths must be excluded, e.g. by restricting to slowly varying paths on which the square-root coefficient ν_1 of Lemma 3.4 stays positive.) Assume also the $B = 0$ anchoring condition: the instantaneous rate implied by the data—the solution of the flow-clearing equation at $t = 0$, with policies at their terminal-implied values—is interior to the band. Then the constrained mass decays no faster than its members find high income, $m_1(t) \ge m_1(0)e^{-\lambda_1 t}$, the flow condition can be solved for the rate, and for $T \le T^*(data, m_1(0))$ an equilibrium transition exists; it is unique in the band under (D1). The horizon T^* shrinks linearly with $m_1(0)$: a thin hand-to-mouth population supports only a thin theorem.*

Proposition 5.3 ($B = 0$ with no constrained population: no local theory)**.** *Suppose $B = 0$ and $m_1 \equiv 0$ on some interval—for instance near $t = 0$ when the initial distribution has no mass at the limit, or following a dispersal episode. On that interval the clearing condition $C(t; r) = Y(t)$ contains the contemporaneous rate* nowhere*: its linearization in the path is a compact operator (as in Theorem 4.3), the constraint is a first-kind equation, and no local solution theory—Picard, Newton, or implicit function—applies. In this regime relaxations are a necessity, not a computational convenience: an ε-elastic bond supply $B_\varepsilon(r) = \varepsilon r$, or a price* *rule $r(t) = F(C(t) - Y(t))$ with F Lipschitz, restores an invertible diagonal $\varepsilon + |\underline{a}|m_1$ and places the problem back in the class of Theorem 5.1; as $\varepsilon \to 0$ one retains only weak limits of approximate equilibria, with clearing recovered in an integrated sense. Existence of exact equilibria in this regime is open—and the neighbouring analysis of Ambrose (2021), in a model where the constrained-mass channel is absent by construction, shows that there the exact problem is generically* unsolvable *while its relaxation is well posed over short horizons.*

The pair of results delivers a sharp economic message about the textbook case $B = 0$: *the hand-to-mouth population is the market.* When it is present, its budget is the margin along which the interest rate clears the market instant by instant, and equilibrium transitions exist. When it is absent, the interest rate is a shadow price with no contemporaneous market to clear, exact instant-by-instant clearing overdetermines the system, and the economically meaningful formulation is one in which either the asset supply has some elasticity or the price responds to imbalances with some friction.

6 Long transitions: monotonicity and multiple self-fulfilling paths

The results above are local in the horizon: the anticipation and history channels each carry a factor T, and the "loop gain" of expectations—beliefs about future rates alter consumption today, which alters the wealth distribution, which alters market-clearing rates tomorrow—scales like T^2. For long horizons, one hopes for global structure. The natural candidate, suggested by the stationary theory, is monotonicity in the rate.

Lemma 6.1 (pathwise comparison fails for debtors)**.** *Couple two economies under rate paths $r_h \geq r_\ell$ (pointwise), identical income realizations, identical initial wealth. Wealth under the higher path dominates wealth under the lower path only if, wherever wealth is negative,*

$$c^{r_\ell}(a,t) - c^{r_h}(a,t) \;\geq\; (r_h(t) - r_\ell(t))\,|a| :$$

the consumption cut induced by higher rates must exceed the added interest burden. For debtors this is a quantitative restriction on the consumption response, not a sign restriction; "higher rates $\Rightarrow$ higher wealth" is not a comparison-principle fact in this model.

In the stationary economy, Achdou et al. (2022) prove the aggregate version of this dominance—saving increasing in r—when the intertemporal elasticity of substitution is at least one ($\gamma \leq 1$) and borrowing is prohibited, and conclude uniqueness of the steady state. Their argument does not survive transplantation to time-varying paths, for identifiable reasons: it perturbs by a *constant* rate (a time-varying perturbation turns the linearized Euler system into a two-point boundary problem with sign-indefinite kernels mixing substitution and wealth effects);

it uses CRRA homogeneity to absorb the rate shift (which requires terminal data covariant under the same scaling—and terminal data drawn from a *different* stationary equilibrium are not); and it aggregates using stationary-distribution identities unavailable along a transition. We therefore state the needed property as a hypothesis and develop its consequences honestly as such:

(H-MONO) For paths $r \le r'$ pointwise in the band: $C(t;r) \ge C(t;r')$ for all t (aggregate consumption falls when the whole rate path rises).

Theorem 6.2 (monotone localization of equilibria)**.** *Let $B > 0$, $S(0) = B$, assume the multiplier map preserves the band and (H-MONO). Then Ψ is antitone, and the two-sided iteration started from the band edges, $\bar{r}^{n+1} = \Psi(\underline{r}^n)$, $\underline{r}^{n+1} = \Psi(\bar{r}^n)$, converges monotonically and uniformly to continuous limits $\underline{r}^* \le \bar{r}^*$ with $\Psi(\underline{r}^*) = \bar{r}^*$ and $\Psi(\bar{r}^*) = \underline{r}^*$.* Every *equilibrium in the band lies between $\underline{r}^*$ and $\bar{r}^*$; any two pointwise-ordered equilibria coincide; and if the limits collapse, $\underline{r}^* = \bar{r}^*$, the equilibrium is unique and the iteration converges to it globally.*

The bracket $[\underline{r}^*, \bar{r}^*]$ is computable: it is the limit of the natural best-response iteration started from the extremes, and the collapse condition can be checked numerically. For $\gamma \le 1$, (H-MONO) is credible—its stationary counterpart holds under a no-borrowing limit by Achdou et al. (2022), and over short horizons up to $O(T)$ by the sensitivity decomposition—but we emphasize it remains unproven for genuinely time-varying perturbations.

For $\gamma > 1$ the situation is qualitatively different.

Proposition 6.3 (risk aversion above one: singular linearizations and multiplicity)**.** *By Höfer (2025, Thm. 1.10), for every $\gamma > 1$ there exist two-state economies (with a no-borrowing limit) and rates $r_1 < r_2 < \rho$ at which stationary aggregate wealth satisfies $S_{\text{stat}}(r_1) > S_{\text{stat}}(r_2)$. Two scope caveats: the construction is existential over economies (a rare-disaster income process), not universal; and it is proved at the no-borrowing limit $\underline{a} = 0$, a boundary case of our maintained $\underline{a} < 0$ (where the hand-to-mouth channel $m_1\underline{a}$ vanishes identically), so its extension to $\underline{a} < 0$ rests on an unproven continuity argument. On the constructed class, and with the bridge from constant paths to transitions understood as a turnpike heuristic:*

(i) (H-MONO) fails in the bulk of long horizons: constant paths at $r_1 < r_2$ order aggregate wealth the wrong way for most of the horizon.

(ii) At a stationary point on the downward-sloping branch, slowly varying path perturbations satisfy $DS[\delta r] \approx S'_{\text{stat}}(r^*)\,\delta r$ with $S'_{\text{stat}}(r^*) \le 0$: the linearized clearing map acquires an approximate kernel (at a fold) or a wrong-signed diagonal. Monotone methods and naïve continuation in the horizon cross a singular linearization; the reduction at a fold is one-dimensional and degenerate—the classical environment for bifurcation of self-fulfilling equilibrium paths, in which a believed future path of rates alters current consumption, reshapes the wealth distribution, and validates the belief.

(iii) When several stationary rates clear the same bond supply, data connecting two of them (initial distribution from one, terminal values from the other; both satisfy $S(0) = B$) admit, on long horizons, transition paths lingering near each steady state as natural candidates for distinct equilibria with identical endpoints*. Under $\gamma \leq 1$ and (H-MONO), by contrast, Theorem 6.2 confines any multiplicity to non-ordered, crossing pairs of paths.*

Remark 6.4 (what is proved and what is conjectured). Unconditional, given the household package: Proposition 4.1, Lemma 4.2, Theorem 4.3, Theorems 5.1 and 5.2, Proposition 5.3. Conditional: Theorem 6.2 under (H-MONO)—credible for $\gamma \leq 1$, refuted in the bulk-of-horizon sense for $\gamma > 1$ on the economies of Höfer (2025, Thm. 1.10). Open: existence for long horizons at $B > 0$ without (H-MONO) (the missing ingredient is an a priori bound keeping equilibrium rate paths in a band); exact uniqueness in the boundary-layer density regime; existence at $B = 0$ without a constrained population; and the rigorous turnpike bridge in Proposition 6.3.

7 Conclusion

The steady-state theory of heterogeneous-agent economies rests on two gifts: the interest rate summarizes the distribution in one number, and impatience pins households to the borrowing constraint, where their behavior is transparent. This paper shows that along a transition both gifts are revoked, and identifies exactly what replaces them. Market clearing in stocks is degenerate—today's rate moves nothing today—and the operative equilibrium condition is the aggregate budget identity in flows, in which the rate's only instantaneous levers are interest on the bond supply and the budgets of hand-to-mouth debtors. Binding at the constraint is governed by terminal conditions rather than impatience, so the hand-to-mouth population—the market's own lever—can disperse endogenously along the path.

Three practical conclusions follow. First, transition equilibria exist and are unique over short horizons under a measurable stability condition, $\theta = |\underline{a}| \sup_t m_1/B < 1$: hand-to-mouth debt below the stock of liquid assets. Second, the textbook case of zero net supply is exactly as well behaved as its constrained population: with no one at the limit, instant-by-instant clearing overdetermines the economy, and slightly elastic supply or a price rule is not an approximation of the true problem but its correct formulation—a conclusion reinforced from the frictionless side by Ambrose (2021). Third, for risk aversion above one, the non-monotone aggregate-saving economies of Höfer (2025) can make the linearized clearing condition singular along a transition, and multiple self-fulfilling paths between the same endpoints become a live possibility: the uniqueness of computed transition dynamics in common calibrations is an article of numerical faith, and testing for it—by restarting path-iteration algorithms from dispersed initial guesses, or by monitoring the sign of the flow-clearing diagonal—seems a worthwhile discipline.

A The household problem: technical package

This appendix collects the assumptions and the analytical results behind Theorem 3.1 and Section 3.1. Throughout, the Hamiltonian is $H(p) = \sup_{c>0}\{u(c) - pc\} = \frac{\gamma}{1-\gamma}p^{1-1/\gamma}$ ($\gamma \neq 1$; $H(p) = -1 - \log p$ for $\gamma = 1$): finite only for $p > 0$ when $\gamma \leq 1$ (for $\gamma > 1$, $H(0) = 0$ but the maximizer escapes), strictly convex, with $H'(p) = -p^{-1/\gamma}$ singular as $p \downarrow 0$ for every γ.

A.1 Maintained assumptions

Assumption 1 (primitives)**.** $\gamma > 0$; $\rho \geq 0$; $0 < y_1 < y_2$; $\lambda_1, \lambda_2 > 0$; $-\infty < \underline{a} < 0$; $T < \infty$; $B \in [0, \infty)$.

Assumption 2 (price band)**.** $r \in \mathcal{K} = \{r \in C([0,T]) : r_{\min} \leq r \leq r_{\max}\}$ with $r_{\max} < y_1/|\underline{a}|$; write $\bar{r} = \max\{|r_{\min}|, r_{\max}\}$ and $\underline{c} = y_1 + r_{\max}\underline{a} > 0$.

Assumption 3 (terminal data)**.** $v_{T,j} \in C^1([\underline{a}, \infty))$, concave, strictly increasing, with constants $c_T^{\min} > 0$, $C_T > 0$ such that *(VT-u)* $\partial_a v_{T,j} \leq u'(c_T^{\min}) =: \bar{p}_T$ and *(VT-l)* $\partial_a v_{T,j}(a) \geq u'(C_T(1 + a - \underline{a}))$ for all a. Both hold for $v_T = v_\infty(\cdot\,; r^*)$ with $r^* \in [r_{\min}, r_{\max}]$, $r^* < \rho$, by the stationary binding equality and asymptotically linear stationary consumption.

Assumption 4 (initial data)**.** $G_0 = (G_{0,1}, G_{0,2})$ nonnegative measures on $[\underline{a}, \infty)$ with total mass one and finite first moment; compact support $[\underline{a}, A_0]$ where stated. Density regimes (D1)/(D1$'$) as in Section 5.

Remark A.1 (not assumed). Neither $\rho > 0$, nor $r(t) < \rho$, nor the stationary finiteness condition $\rho > (1-\gamma)r$ appears in this appendix. On a finite horizon the terminal valuation replaces transversality.

A.2 Value function and constrained viscosity solutions

The value function is $v_j(a,t) = \sup \mathbb{E}[\int_t^T e^{-\rho(s-t)}u(c_s)ds + e^{-\rho(T-t)}v_{T,y_T}(a_T)]$ over progressively measurable $c \geq 0$ maintaining $a_s \geq \underline{a}$; admissibility is nonempty uniformly over $\mathcal{K}$ by Assumption 2 (hold the state consuming $\underline{c}_{y_s}(s) > 0$). Standard arguments give pathwise wealth and budget bounds, finiteness and linear growth of v, concavity and strict monotonicity in a, and dynamic programming.

Definition A.2 (constrained viscosity solution)**.** (v_1, v_2), continuous on $[\underline{a}, \infty) \times [0, T]$, is a constrained viscosity solution if it is a viscosity subsolution of (1) on $(\underline{a}, \infty) \times [0, T)$, a supersolution on $[\underline{a}, \infty) \times [0, T)$ (boundary included), and matches the terminal data. For concave solutions the supersolution property at $\underline{a}$ is equivalent to (2) (Soner, 1986a; Capuzzo-Dolcetta and Lions, 1990).

Theorem A.3 (existence, comparison, uniqueness)**.** *Under Assumptions 1–3, for every $r \in \mathcal{K}$ the value function is the unique constrained viscosity solution in the class of continuous functions concave, nondecreasing, of linear growth in a.*

Proof architecture. (i) Sub/supersolution properties from dynamic programming; the state-constraint supersolution property at $\underline{a}$ via Soner's inward cone, whose hypothesis is exactly $\underline{c}_j(t) \geq \underline{c} > 0$, uniform over $\mathcal{K}$ (Soner, 1986a,b; Capuzzo-Dolcetta and Lions, 1990). (ii) The singular Hamiltonian is handled by truncation: comparison is proved for a globally Lipschitz $H_\flat$ agreeing with H on the compact gradient range delivered by Theorems A.5–A.6; solutions never see the truncation. This interface between the gradient bounds and the comparison argument must be made explicit, since the range is not free in finite horizon. (iii) Comparison for the weakly coupled system: the switching coupling is quasi-monotone, so the system comparison principles of Ishii and Koike (1991) and Engler and Lenhart (1991) combine with parabolic doubling (Crandall, Ishii, and Lions, 1992) and a linear penalization at infinity; the boundary is handled on the supersolution side by the inward cone, uniformly in t. (iv) Verification along Filippov solutions of the closed-loop dynamics (Appendix B). (v) C^1 regularity in a: a concave viscosity solution cannot kink where the Hamiltonian is strictly convex (Cannarsa and Sinestrari, 2004); switching terms are zeroth order and harmless. □

Remark A.4 (terminal boundary layer)*.* The boundary inequality (2) holds at every $t < T$ even if the terminal data violate it at $t = T$: a marginal unit of wealth at the limit can always be consumed immediately at rate $\geq \underline{c}_j(t)$. If $\partial_a v_{T,j}(\underline{a}) < u'(\underline{c}_j(T))$, the boundary gradient jumps as $t \uparrow T$—a terminal boundary layer in $\partial_a v$, invisible in v.

A.3 Gradient bounds by barriers

Formally, $w_j = \partial_a v_j$ solves the cooperative transport system

$$\partial_t w_j + s_j \, \partial_a w_j = (\rho + \lambda_j - r(t)) \, w_j - \lambda_j w_{-j}, \qquad w_j(\cdot, T) = v'_{T,j}; \tag{6}$$

the rigorous versions of the following run the barriers on vanishing-viscosity or implicit finite-difference approximations and pass to the limit.

Theorem A.5 (upper bound)**.** *For all j, a, t and $r \in \mathcal{K}$: $\partial_a v_j(a, t) \leq p_{\max} := \max\{\bar{p}_T, u'(\underline{c})\} \, e^{(r_{\max} - \rho)^+ T}$.*

Sketch. $W(t) = \max\{\bar{p}_T, u'(\underline{c})\} \exp(\int_t^T (r - \rho)^+)$ is a spatially constant supersolution of (6) (the required inequality $-(r - \rho)^+ W \leq (\rho - r)W$ always holds); at the boundary the state-constraint mechanism can raise $w_j(\underline{a}, t)$ at most to the obstacle $u'(\underline{c}_j(t)) \leq u'(\underline{c}) \leq W(t)$; at T, $W \geq \bar{p}_T$. Cooperative comparison concludes. What replaced the stationary shortcut: stationarily $w_1(\underline{a}) = u'(y_1 + r\underline{a})$ exactly (binding under $r < \rho$); here binding may fail (Lemma 3.2), the bound comes from the obstacle or the terminal data, and the cap (VT-u) is a genuine assumption. □

Theorem A.6 (lower bound; linear consumption growth)**.** *There are $\kappa > 0$ and $C = C(\gamma, y_2, \bar{r}, \rho, \lambda)$ with $\partial_a v_j(a,t) \geq \kappa e^{-CT}(1+a-\underline{a})^{-\gamma}$, equivalently $c_j(a,t) \leq \kappa^{-1/\gamma} e^{CT/\gamma}(1+a-\underline{a})$, uniformly over $\mathcal{K}$.*

Sketch. The ansatz $m(a,t) = \delta(t)(1 + a - \underline{a})^{-\gamma}$, identical across income states so the switching coupling cancels, is a subsolution of (6) for $\delta(t) = \kappa e^{-C(T-t)}$, using the one-sided drift bound $s_j \leq y_2 + \bar{r}|\underline{a}| + r^+_{\max}(1 + a - \underline{a})$; at T it sits below $v'_{T,j}$ by (VT-l); at the boundary below the obstacle. □

A.4 Binding, dispersal, and the square-root law

Lemma 3.2 follows from the stability of v near T plus concavity: as $t \uparrow T$, $\partial_a v_1(\underline{a}^+, t) \to \max\{\partial_a v_{T,1}(\underline{a}), u'(\underline{c}_1(T))\}$, and comparing the implied desired consumption with hand-to-mouth income $\underline{c}_1(t)$ yields the two cases; Corollary 3.3 is the specialization to canonical terminal data.

Example A.7 (closed form: binding fails although $r < \rho$)**.** Remove income risk ($\lambda_1 = \lambda_2 = 0$, income y), let $r < \rho$ be constant, and take terminal data $v_T(a) = \omega_T^{\gamma} u(a+h_T)$ with $h_T > -\underline{a}$. The problem solves in closed form: $c(a,t) = (a + h(t))/\omega(t)$ with $h(t) = \int_t^T y\, e^{-r(s-t)} ds + h_T e^{-r(T-t)}$ (human wealth) and $\dot{\omega} = \frac{\rho-(1-\gamma)r}{\gamma}\omega - 1$, $\omega(T) = \omega_T$ (the inverse marginal propensity to consume). The boundary saving rate $s(\underline{a}, t) = y + r\underline{a} - (\underline{a} + h(t))/\omega(t)$ is strictly positive at $t = T$ for ω_T large (a patient, high-valuation terminal datum): binding fails on a whole terminal interval despite $r < \rho$. As $T - t \to \infty$, h and ω approach their stationary limits and binding is restored: the finite-horizon boundary behavior agrees with the stationary one only at distance $O(1)$ from T (a turnpike statement).

On Lemma 3.4. Matching orders in (6) at the boundary along a binding interval, with $w_1(\underline{a}, t) = u'(\underline{c}_1(t))$ so that $\partial_t w_1(\underline{a}, t) = u''(\underline{c}_1(t))\, \dot{r}(t)\underline{a}$, yields the displayed $\nu_1(t)$; the singular products cancel exactly as in the stationary case. The sign $\nu_1(t) \geq 0$ coincides with the complementarity (KKT) rate of the state-constraint multiplier, so the square-root law and binding degrade together. A rigorous version freezes coefficients on short time intervals and compares with sub- and super-boundary layers; we flag it as the delicate step of the package.

A.5 Stability in the rate path

Lemma A.8 (continuity and Lipschitz stability)**.** *If $r_n \to r$ uniformly in $\mathcal{K}$, then $v^{r_n} \to v^r$ locally uniformly (half-relaxed limits plus comparison, the boundary obstacle converging uniformly by Assumption 2), and policies converge locally uniformly on $[\underline{a}, \infty) \times [0, T)$. Moreover*

$$|v_j^r(a,t) - v_j^{r'}(a,t)| \;\leq\; C^\star(1 + a - \underline{a})\, \|r - r'\|_{C([t,T])},$$

with $C^\star$ uniform over $\mathcal{K}$; only the future of the path enters.

Remark A.9 (policies across the boundary layer). Pointwise Lipschitz stability of policies in the rate fails near the limit: the square-root layer's coefficient and its binding/release times move with r, making $\|c^r - c^{r'}\|_\infty \lesssim \|r - r'\|^{1/2}$ the honest generic rate. The equilibrium analysis needs less: in aggregates, the atom's consumption $y_1 + r(t)\underline{a}$ is exactly Lipschitz with constant $|\underline{a}|$, and the interior part meets the layer only on a set of small mass ($O(\delta^{3/2})$ under (D1), $O(\delta)$ under (D1′)); the pointwise loss integrates away, up to a logarithm under (D1′). This atom/interior division is the interface used in Section 5.

B Distributional dynamics

Definition B.1 (measure-valued forward solutions)**.** Given policies c_j and drifts s_j, a narrowly continuous family $t \mapsto (G_{1,t}, G_{2,t})$ of nonnegative measures on $[\underline{a}, \infty)$ solves the forward equation if for all $\varphi_j \in C_b^1([\underline{a}, \infty) \times [0, T])$—including test functions *not* vanishing at $\underline{a}$—

$$\frac{d}{dt} \sum_j \int \varphi_j \, dG_{j,t} = \sum_j \int \Big[\partial_t \varphi_j + s_j \partial_a \varphi_j + \lambda_j(\varphi_{-j} - \varphi_j)\Big] \, dG_{j,t}, \qquad G_{j,0} = G_{0,j}.$$

This is the time-dependent version of the extended weak formulation of Achdou et al. (2022) (Supplementary Appendix E), written out there for the two-type equation in stationary form with the time-dependent case declared analogous; unbounded test functions such as $\varphi = a$ are admitted by cutoff for solutions with uniformly bounded first moments.

Lemma B.2 (one-sided Lipschitz drift)**.** *Since $c_j(\cdot, t)$ is nondecreasing, $(s_j(a,t) - s_j(b,t))(a - b) \le \bar{r}|a - b|^2$: the drift is one-sided Lipschitz uniformly in t and $\mathcal{K}$, despite the square-root singularity (which has the contractive sign); and $s_1(\underline{a}, t) \ge 0$ always.*

Theorem B.3 (well-posedness of the forward equation)**.** *There is a unique measure-valued solution: the law of the piecewise-deterministic process alternating the forward Filippov flows of $\dot{a} = s_j(a,t)$ with the Poisson switching clock. Trajectories merge at $\underline{a}$ (atom formation) but never cross; uniqueness follows by duality in the one-dimensional one-sided-Lipschitz transport theory (Filippov, 1960; Bouchut and James, 1998; Poupaud and Rascle, 1997), extended to the switching system by Duhamel. Compact supports propagate with explicit bounds, and the first moment $S(t)$ is Lipschitz in t uniformly over $\mathcal{K}$—the rigorous content of the flow identity in Lemma 4.2.*

Proposition B.4 (atom dynamics: growth and discontinuous release)**.** *Let $m_1(t) = G_{1,t}(\{\underline{a}\})$. On binding intervals m_1 is absolutely continuous with*

$$\dot{m}_1(t) = \Phi_1(t) - \lambda_1 m_1(t), \qquad \Phi_1(t) = -\lim_{a \downarrow \underline{a}} s_1(a,t) g_1(a,t) \; \ge 0,$$

inflow through the square-root drift feeding the mass, leakage only through switching to high *income. At a release time t_0 (binding fails just after t_0, e.g. the rate crosses below the terminal-implicit rate of Corollary 3.3), the mass detaches* as a whole *and travels into the interior along the Filippov characteristic through $(\underline{a}, t_0)$, decaying at rate λ_1; hence m_1 drops to zero discontinuously, and re-accumulates if binding resumes. In particular $t \mapsto m_1(t)$ is of bounded variation but not continuous, and the contemporaneous clearing sensitivity $B + |\underline{a}|m_1(t)$ of Section 4 is switched on and off by the binding analysis of Section 3.1.*

C Proof architectures for Sections 4–6

Lemma 4.2. Take $\varphi_j = a$ in Definition B.1 (justified by cutoff): the switching terms cancel, leaving $\dot{S} = \sum_j \int s_j \, dG_{j,t} = Y + rS - C$. Given $S(0) = B$ and $C = Y + rB$, $\dot{S} = r(S - B)$, so $S \equiv B$ by Grönwall; conversely $S \equiv B$ forces $\dot{S} = 0$, i.e. $C = Y + rB$.

Theorem 4.3. (i) is Proposition 4.1. (ii): the value at dates $t \leq t_0 - h$ responds to a perturbation supported on $(t_0 - h, t_0 + h)$ only through a Duhamel source active on a window of width $2h$ (Lemma A.8); the distribution responds directly only during the window (drift term $a \, \delta r$); integrating the policy response against the distribution over $[0, t_0]$ gives the bound, with the logarithmic factor under (D1′) inherited from Remark A.9. (iii): (ii) plus Arzelà–Ascoli; the first-kind character is the vanishing diagonal.

Sensitivity decomposition and (ANCH). Write $B[\Psi(r)(t) - \Psi(r')(t)] = \int (c^r - c^{r'})(\cdot, t) \, dG_t^{r'} + \int c^r(\cdot, t) \, d(G_t^r - G_t^{r'})$. The first term splits into the atom—where $c^r - c^{r'} = \underline{a}\,(r(t) - r'(t))$, contributing $\theta\eta$—and the interior, where marginal values live in a compact range and Lemma A.8 gives $C_1(T - t)\eta$ (the anticipation channel). The second term is controlled by the W_∞-stability of the Filippov flow (one-sided Lipschitz constant $\bar{r}$), $W_\infty(G_t^r, G_t^{r'}) \leq L_2 t\eta$, integrated against a consumption function Lipschitz away from the limit and $\frac{1}{2}$-Hölder near it: Lipschitz transfer under (D1), log-Lipschitz under (D1′) by splitting the quantile axis at the mismatch scale (mass exchanged between atom and layer moves consumption only at second order because consumption is continuous across the limit). As $T \to 0$, policies converge to their terminal-implied values and $\Psi(r)(t)$ converges, uniformly over $\mathcal{K}$, to the affine map $r(t) \mapsto \big[\int_{(\underline{a}, \infty)} c_T \, dG_0 + m_1(0) \min\{c_{T,1}(\underline{a}), y_1 + r(t)\underline{a}\} - Y(0) \big] / B$, whose fixed point r^{**} is unique when $\theta_0 = m_1(0)|\underline{a}|/B < 1$; **(ANCH)** requires $r^{**} \in (r_{\min} + \delta, r_{\max} - \delta)$ and delivers band invariance of Ψ for small T.

Theorem 5.1. (i) Schauder on the compact convex hull of $\Psi(\mathcal{K})$ inside the band. (ii) Banach with the (D1) decomposition. (iii) Inserting $\eta = \|r - r'\|$ into the (D1′) decomposition gives $1 \leq \bar{\theta} + C_1 T + C_2 T(1 + \log_+(1/\eta))$ for $\eta > 0$, i.e. $\eta \leq \exp(-(1 - \bar{\theta} - (C_1 + C_2)T)/(C_2 T))$; the log-Lipschitz estimate does not self-improve to zero, and the two-directional time coupling (anticipation backward, history forward) blocks a Grönwall–Osgood closure.

Proposition C.1 (diagonal elimination and its price). *Solving the flow condition exactly for $r(t)$ yields*

$$r(t) = \widetilde{\Psi}(r)(t) = \frac{C^{\mathrm{int}}(t;r) + m_1(t;r)\, y_1 - Y(t)}{B + m_1(t;r)\, |\underline{a}|},$$

with C^{int} the consumption of the unconstrained. $\widetilde{\Psi}$ has zero contemporaneous sensitivity and a uniformly invertible denominator for $B > 0$; under (D1), Theorem 5.1(ii) holds for $\widetilde{\Psi}$ without the hypothesis $\theta < 1$. Under (D1′), the separated atom mass is only $\frac{1}{2}$-Hölder stable in the path ($|m_1(t;r) - m_1(t;r')| \le C\sqrt{T\eta}$, the mismatch mass between atom and boundary layer), and the loss enters $\widetilde{\Psi}$ at order one: no contraction. The trade-off is sharp at the level of these estimates: aggregates are log-Lipschitz because mass exchanged between atom and layer moves consumption at second order; any map separating the atom pays the Hölder price. Under (D1′) the economic threshold $\theta < 1$ is genuinely load-bearing.

Theorem 5.2. Under binding on $[0,T]$, atom outflow is only through switching, giving $m_1(t) \ge m_1(0)e^{-\lambda_1 t}$; the map $\widetilde{\Psi}_0(r)(t) = \big(C^{\mathrm{int}}(t;r) + m_1(t;r)y_1 - Y(t)\big)/\big(|\underline{a}|m_1(t;r)\big)$ is then well defined, its denominator bounded below by $|\underline{a}|m_1(0)e^{-\lambda_1 T}$; the Schauder and contraction steps proceed as in Theorem 5.1, with all constants scaled by $1/(|\underline{a}|m_1(0))$—whence T^* degrading linearly in $m_1(0)$. Under (D1′), two equilibria satisfy $\eta \le C\sqrt{T\eta}$, i.e. $\eta \le CT$; exact uniqueness is open.

Theorem 6.2. Standard mixed-monotone iteration for antitone maps on the order-complete cone of $C([0,T])$: (H-MONO) makes Ψ antitone, so the two-sided scheme generates nested order intervals; compactness of Ψ's range upgrades monotone pointwise convergence to uniform convergence with continuous limits; any equilibrium is squeezed by induction, and ordered equilibria coincide by antitonicity. A degree-theoretic continuation in T from the small-horizon equilibrium (index $+1$) extends existence along the branch while a priori interiority holds and $I - D\Psi$ stays invertible; (H-MONO) supplies sign structure but not spectral control, so this continuation remains conditional—and Proposition 6.3(ii) identifies exactly where it fails for $\gamma > 1$.

References

Açıkgöz, Ömer T. 2018. "On the Existence and Uniqueness of Stationary Equilibrium in Bewley Economies with Production." *Journal of Economic Theory* 173: 18–55.

Achdou, Yves, Francisco J. Buera, Jean-Michel Lasry, Pierre-Louis Lions, and Benjamin Moll.

2014. “Partial Differential Equation Models in Macroeconomics.” *Philosophical Transactions of the Royal Society A* 372: 20130397.

Achdou, Yves, Jiequn Han, Jean-Michel Lasry, Pierre-Louis Lions, and Benjamin Moll. 2022. “Income and Wealth Distribution in Macroeconomics: A Continuous-Time Approach.” *Review of Economic Studies* 89 (1): 45–86.

Aiyagari, S. Rao. 1994. “Uninsured Idiosyncratic Risk and Aggregate Saving.” *Quarterly Journal of Economics* 109 (3): 659–684.

Ambrose, David M. 2021. “Existence Theory for a Time-Dependent Mean Field Games Model of Household Wealth.” *Applied Mathematics & Optimization* 83 (3): 2051–2081.

Auclert, Adrien, Bence Bardóczy, Matthew Rognlie, and Ludwig Straub. 2021. “Using the Sequence-Space Jacobian to Solve and Estimate Heterogeneous-Agent Models.” *Econometrica* 89 (5): 2375–2408.

Bayer, Christian, Alan D. Rendall, and Klaus Wälde. 2019. “The Invariant Distribution of Wealth and Employment Status in a Small Open Economy with Precautionary Savings.” *Journal of Mathematical Economics* 85: 17–37.

Boppart, Timo, Per Krusell, and Kurt Mitman. 2018. “Exploiting MIT Shocks in Heterogeneous-Agent Economies: The Impulse Response as a Numerical Derivative.” *Journal of Economic Dynamics and Control* 89: 68–92.

Bouchut, François, and François James. 1998. “One-Dimensional Transport Equations with Discontinuous Coefficients.” *Nonlinear Analysis* 32 (7): 891–933.

Cannarsa, Piermarco, and Rossana Capuani. 2018. “Existence and Uniqueness for Mean Field Games with State Constraints.” In *PDE Models for Multi-Agent Phenomena*, Springer INdAM Series 28: 49–71.

Cannarsa, Piermarco, Rossana Capuani, and Pierre Cardaliaguet. 2021. “Mean Field Games with State Constraints: From Mild to Pointwise Solutions of the PDE System.” *Calculus of Variations and Partial Differential Equations* 60: article 108.

Cannarsa, Piermarco, and Carlo Sinestrari. 2004. *Semiconcave Functions, Hamilton–Jacobi Equations, and Optimal Control.* Boston: Birkhäuser.

Capuzzo-Dolcetta, Italo, and Pierre-Louis Lions. 1990. “Hamilton–Jacobi Equations with State Constraints.” *Transactions of the American Mathematical Society* 318 (2): 643–683.

Cardaliaguet, Pierre, and Charles-Albert Lehalle. 2018. “Mean Field Game of Controls and an Application to Trade Crowding.” *Mathematics and Financial Economics* 12: 335–363.

Chen, Cangxiong, Sigmund Ellingsrud, Fabian Harang, Alfonso Irarrazabal, and Avi Mayorcas. 2025. "Stochastic Analysis of Overlapping Generations Models under Incomplete Markets." arXiv:2509.05170.

Cirant, Marco, Roberto Gianni, and Paola Mannucci. 2020. "Short-Time Existence for a General Backward–Forward Parabolic System Arising from Mean-Field Games." *Dynamic Games and Applications* 10 (1): 100–119.

Crandall, Michael G., Hitoshi Ishii, and Pierre-Louis Lions. 1992. "User's Guide to Viscosity Solutions of Second Order Partial Differential Equations." *Bulletin of the American Mathematical Society* 27 (1): 1–67.

Engler, Hans, and Suzanne M. Lenhart. 1991. "Viscosity Solutions for Weakly Coupled Systems of Hamilton–Jacobi Equations." *Proceedings of the London Mathematical Society* 63 (1): 212–240.

Filippov, Aleksei F. 1960. "Differential Equations with Discontinuous Right-Hand Side." *Matematicheskii Sbornik* 51: 99–128. English translation: *AMS Translations, Series 2* 42 (1964): 199–231.

Fujii, Masaaki, and Akihiko Takahashi. 2022. "A Mean Field Game Approach to Equilibrium Pricing with Market Clearing Condition." *SIAM Journal on Control and Optimization* 60 (1): 259–279.

Gomes, Diogo A., and João Saúde. 2021. "A Mean-Field Game Approach to Price Formation." *Dynamic Games and Applications* 11 (1): 29–53.

Graber, P. Jameson, and Alain Bensoussan. 2018. "Existence and Uniqueness of Solutions for Bertrand and Cournot Mean Field Games." *Applied Mathematics & Optimization* 77 (1): 47–71.

Graber, P. Jameson, and Sergio Mayorga. 2021. "A Note on Deterministic Mean Field Games of Controls with State Constraints: Existence of Mild Solutions." arXiv:2109.11655 (revised 2023).

Höfer, F. 2025. "Stationary Heterogeneous-Agent Models in Continuous Time." arXiv:2510.26065 (revised 2026).

Huggett, Mark. 1993. "The Risk-Free Rate in Heterogeneous-Agent Incomplete-Insurance Economies." *Journal of Economic Dynamics and Control* 17 (5–6): 953–969.

Ishii, Hitoshi, and Shigeaki Koike. 1991. "Viscosity Solutions for Monotone Systems of Second-Order Elliptic PDEs." *Communications in Partial Differential Equations* 16 (6–7): 1095–1128.

Kaplan, Greg, Benjamin Moll, and Giovanni L. Violante. 2018. "Monetary Policy According to HANK." *American Economic Review* 108 (3): 697–743.

Kobeissi, Ziad. 2022. "Mean Field Games with Monotonous Interactions through the Law of States and Controls of the Agents." *NoDEA Nonlinear Differential Equations and Applications* 29: article 52.

Porretta, Alessio. 2014. "On the Planning Problem for the Mean Field Games System." *Dynamic Games and Applications* 4 (2): 231–256.

Poupaud, Frédéric, and Michel Rascle. 1997. "Measure Solutions to the Linear Multi-Dimensional Transport Equation with Non-Smooth Coefficients." *Communications in Partial Differential Equations* 22 (1–2): 337–358.

Shigeta, Yuki. 2023. "Existence of Invariant Measure and Stationary Equilibrium in a Continuous-Time One-Asset Aiyagari Model: A Case of Regular Controls under Markov Chain Uncertainty." SSRN Working Paper 4510345; companion paper SSRN 4510320.

Soner, Halil Mete. 1986a. "Optimal Control with State-Space Constraint I." *SIAM Journal on Control and Optimization* 24 (3): 552–561.

Soner, Halil Mete. 1986b. "Optimal Control with State-Space Constraint II." *SIAM Journal on Control and Optimization* 24 (6): 1110–1122.